Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18RA71]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

FIELD AND SERVICE ROBOTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Explain the recent developments in Field Robotics Technology.	08
	b.	Classify the Assistive Robotics based on their applications.	06
2.	a.	Explain the general robot localization problem and the associated control strategy.	09
	b.	With suitable sketches, explain the features of positioning beacon systems.	05
3.	a.	Discuss in detail, the Visibility graph and Voronoi diagram approaches of path planning.	10
	b.	Draw the block diagram of incremental planning scheme for path planning in mobile robots.	04
		OR	
4.	a.	List out the drawbacks of traditional methods of path planning.	06
	b.	With a suitable technique, explain the obstacle avoidance in mobile Robots.	08
5.	a.	Discuss the applications of Robots in space exploration.	07
	b.	Explain the features of Exoskeleton Robots and list their applications.	07
		OR	
6.	a.	What are the applications of field robots in military applications?	06
	b.	Discuss the following in detail: i. Applications of Aerial Robotics ii. Features of Underwater Robotics	08
7.	a.	Explain the following terms with respect to Mobile Robots: i. Stability ii. Manoeuvrability iii. Controllability	09
	b.	Explain with a neat diagram about Slip/Skid Locomotion.	05
		OR	
8.	a.	Explain any two types of legged Robot Locomotion with suitable examples.	06
	b.	Draw the block diagram of a machine vision system and explain the components.	08